



Cybersecurity Department

Protecting the digital foundation of railway infrastructure

Our mission

Cybersecurity as an integral part of railway infrastructure governance

Protect the shared digital infrastructure that supports railway operations



Critical infrastructure

Security of the shared infrastructure perimeter



National coordination

Communication with the Czech cybersecurity authority



Embedded governance

Cybersecurity is part of Railway Telematics

Management perspective: cybersecurity protects operational continuity, trust and safe modernization.

Security Transformation Portfolio

Identity & Access Security

- Multi-Factor Authentication (MFA)
- Privileged Access Management (PAM)
- Identity Management (IDM)

Network & Perimeter Protection

- Next Generation Firewall (NGFW)
- Network Segmentation
- new DMZ Architecture
- DDoS Protection

Threat Detection & Response

- Security Information and Event Management (SIEM)
- Extended Detection & Response (XDR)

Data Protection

- Data Loss Prevention (DLP)

Strategic Objective Reduce cyber risk, increase operational resilience, and establish a modern security architecture for both IT and operational technology environments.

Outcome: stronger identity security, improved threat detection, enhanced infrastructure resilience, and protection of critical railway services.

Operating model

Three complementary capabilities under a single cybersecurity function

01

Security Architecture

- Security requirements for projects
- Implementation of security controls
- IT, communications and telecom standards

02

Governance & Risk

- Internal security testing
- Corrective measures and assurance
- ISMS, asset and risk management

03

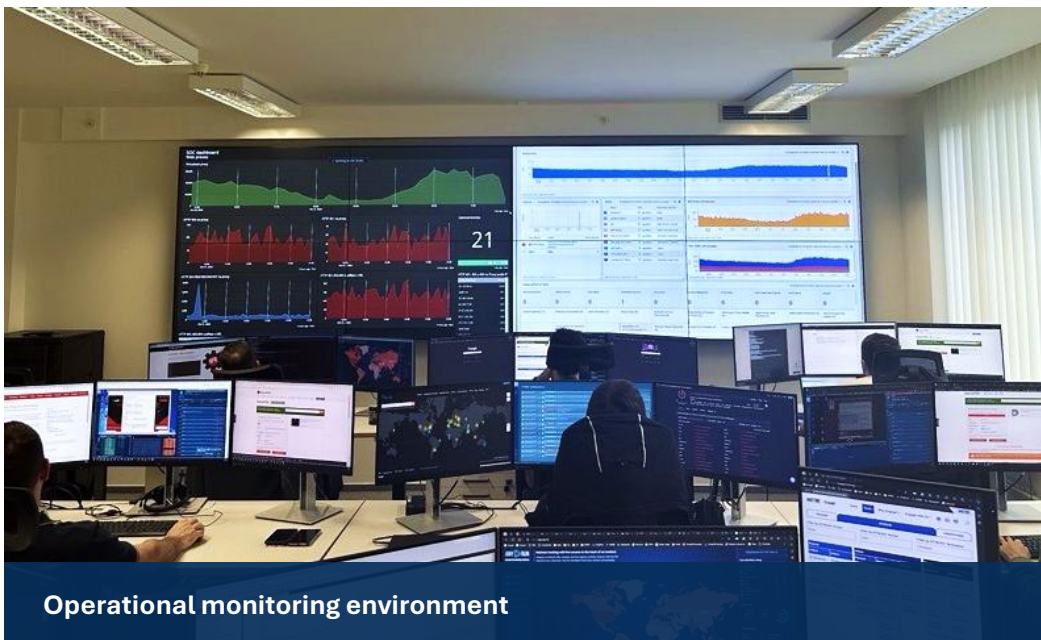
SOC & Incident Response

- Security monitoring
- Incident handling and penetration testing
- Vulnerability analysis and access approval

Clear separation of design, assurance and operations enables independent control while maintaining fast coordination.

24/7 Security Operations Centre

Continuous monitoring and coordinated incident response across shared infrastructure



24/7

service

6

monitoring tools

TF-CSIRT

listed member

Core responsibilities

- Detect attacks and malicious activity
- Analyse vulnerabilities and compromised systems
- Coordinate and resolve security incidents

Priorities for a railway infrastructure manager

Strengthening cyber resilience across enterprise IT and operational technology

1

Secure-by-design investment

Embed security requirements in construction, technology and digitalisation projects.

2

IT/OT risk convergence

Extend governance, visibility and controls to operational technology environments.

3

Operational resilience

Use continuous monitoring, incident response and recovery preparedness to protect service continuity.

4

Regulatory readiness

Keep policies, standards and methodologies aligned with cybersecurity obligations.

Potential exchange topics: railway IT/OT monitoring, incident cooperation, supply-chain security and resilience exercises, vibecoding governance, drone security

Thank you

time for your questions..